HYPOTHESIS TESTING

Hypothesis: An idea or explanation for something that is based on known facts but has not yet been proven to be true.

In Statistics :

A hypothesis is a statement or assumption about a population parameter, such as:

Population mean → μ

Population proportion → p

Population variance → σ^2

Since it is usually difficult to collect data from the entire population, we take a sample data to test the hypothesis.

What is Hypothesis Testing?

Hypothesis testing is a statistical method used to determine whether there is enough evidence in sample data to reject a claim or assumption about a population.

Population and Sample

Population : The entire group we are interested in studying.

Sample : A subset of the population selected for analysis.

Suppose a company produces 1,00,000 tea packets.

(It is difficult to check every packet . So we select 30 packets)

Population=1,00,000 packets

Sample=30 packets

We use the sample to make an inference about the population.

Why Do We Need Hypothesis Testing?

Average packet weight = 500 g

randomly select 30 packets and calculate: $\bar{x} = 495g$

Does this mean that the company's claim is wrong? **Not necessarily.**

The sample mean may differ from the population mean simply because of **sampling variation**.

For example:

- Sample 1 → 495 g
- Sample 2 → 502 g
- Sample 3 → 498 g
- Sample 4 → 501 g

Different samples can produce different means.

Therefore, we need a statistical method to determine whether the observed difference is **statistically significant**.

That method is called **Hypothesis Testing**.

Types of Hypotheses

There are two main hypotheses:

1. Null Hypothesis
2. Alternative Hypothesis

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1. **Null Hypothesis (H_0)**

   The **Null Hypothesis** represents the statement of no difference, or no effect. It is assumed to be true until proven otherwise

2. **Alternative Hypothesis (H_a or H_1)**

   The **Alternative Hypothesis** represents the claim we want to investigate or find evidence for.

Example:

1. A company claims:Average tea packet weight is 500 g.

   H0 : μ=500

2. If we want to check whether the average weight is different from 500 g:

$$H_a : \mu \neq 500$$

Types of Alternative Hypothesis

1. **Two-Tailed Test**

   Used when we want to check whether the population parameter is **different** from a specific value.

   Example

   Company claims: $\mu = 500$
   We want to check whether the actual mean is different.

$$H_0 : \mu = 500$$
$$H_a : \mu \neq 500$$

   The difference could be: Greater than 500 or Less than 500

   Therefore, it is called a **two-tailed test**.

2. **Right-Tailed Test**

   Used when we want to check whether the population parameter is **greater than** a specific value.

Example

A company claims the average battery life is 10 hours.

We want to check whether it is actually **greater than 10 hours**.

$$H_0: \mu = 10$$
$$H_a: \mu > 10$$

This is a **right-tailed test**.

3. **Left-Tailed Test**

Used when we want to check whether the population parameter is **less than** a specific value.

Example

A company claims that the average tea packet weight is 500 g.

We want to check whether the actual average is **less than 500 g**.

$$H_0: \mu = 500$$
$$H_a: \mu < 500$$

This is a **left-tailed test**.

Significance Level (α)

The **significance level** is the maximum probability of rejecting the null hypothesis when it is actually true.

Significance Level	**Meaning**
0.10	10%
0.05	5%
0.01	1%

Test Statistic

A **test statistic** is a numerical value calculated from sample data.
It tells us how far our sample result is from the hypothesized population value.

For a Z-test:

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

For a one-sample t-test:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Where:

- $\bar{x}$ = Sample mean
- μ_0 = Hypothesized population mean
- σ = Population standard deviation
- s = Sample standard deviation
- n = Sample size

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p-value

The **p-value** helps us determine whether the sample evidence is strong enough to reject the null hypothesis.

Decision Rule

If:

$$p\text{-value} \leq \alpha$$

Then:

$$\boxed{\text{Reject } H_0}$$

If:

$$p\text{-value} > \alpha$$

Then:

$$\boxed{\text{Fail to Reject } H_0}$$

Important

We generally say : **Fail to reject H_0** instead of : Accept H_0

Because hypothesis testing does not prove that the null hypothesis is definitely true.

Critical Value

A **critical value** is a boundary that separates the rejection region from the non-rejection region.

For a Z-test at:

$$\alpha = 0.05$$

Two-tailed test

$$Z_{critical} = \pm 1.96$$

Right-tailed test

$$Z_{critical} = 1.645$$

Left-tailed test

$$Z_{critical} = -1.645$$

Rejection Region

The **rejection region** is the range of test statistic values where we reject H_0.

Two-tailed test

$$Z < -1.96$$

or

$$Z > 1.96$$

Right-tailed test

$$Z > 1.645$$

Left-tailed test

$$Z < -1.645$$

Steps of Hypothesis Testing

Step 1: State the Hypotheses

Step 2: Choose Significance Level → $\alpha = 0.05$

Step 3: Select the Appropriate Statistical Test

Z-test , t-test , Chi-square test , ANOVA

Step 4: Calculate the Test Statistic

Step 5: Calculate the p-value

Step 6: Make a Decision

If:

$$p \leq \alpha$$

Reject H_0.

If:

$$p > \alpha$$

Fail to reject H_0.

Step 7: State the Conclusion in Context

Type I and Type II Errors

Hypothesis testing can result in two types of errors.

Type I Error

Rejecting H_0 when H_0 is actually true.

$$\boxed{\text{False Positive}}$$

Probability of Type I error:

$$P(\text{Type I Error}) = \alpha$$

If:

$$\alpha = 0.05$$

There is a 5% significance level for Type I error.

Type II Error

Failing to reject H_0 when H_0 is actually false.

$$\boxed{\text{False Negative}}$$

Probability of Type II error:

$$P(\text{Type II Error}) = \beta$$

Z-TEST

1. **What is a Z-test?**

   A Z-test is a statistical hypothesis test used to determine whether a sample mean is significantly different from a hypothesized population mean when the population standard deviation is known.

   The classic one-sample Z-test for a mean is:

$$\boxed{Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}}$$

   Where:

   - $\bar{x}$ = Sample mean
   - μ_0 = Hypothesized population mean
   - σ = Population standard deviation
   - n = Sample size

   When to Use a Z-Test?

   A Z-test for a population mean is appropriate when:

   1. The population standard deviation σ is known, and
   2. The population is normally distributed, or the sample size is sufficiently large for the Central Limit Theorem to justify the normal approximation.

Example

Problem

A tea company claims that the average weight of its tea packets is **500 g**.

The population standard deviation is known to be:

$$\sigma = 10g$$

A sample of:

$$n = 30$$

packets has a sample mean:

$$\bar{x} = 495g$$

Test whether the average weight is **less than 500 g** at a 5% significance level.

Step 1: State the Hypotheses

The claim is:

$$\mu = 500$$

We want to check whether the mean is less than 500.

Therefore:

$$H_0 : \mu = 500$$

$$H_a : \mu < 500$$

This is a **left-tailed test**.

Step 2: Significance Level

$$\alpha = 0.05$$

Step 3: Calculate Z-statistic

Formula:

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Substitute:

$$Z = \frac{495 - 500}{10 / \sqrt{30}}$$

$$Z \approx -2.74$$

Step 4: Critical Value

For a left-tailed test at:

$$\alpha = 0.05$$

The critical value is:

$$Z_{critical} = -1.645$$

Step 5: Compare

Our test statistic:

$$Z = -2.74$$

Critical value:

$$Z_{critical} = -1.645$$

Since:

$$-2.74 < -1.645$$

we reject H_0.

Step 6: Conclusion

> At the 5% significance level, there is sufficient statistical evidence to conclude that the average weight of the tea packets is significantly less than 500 g.

T-TEST

What is a T-test?

A **t-test** is a statistical hypothesis test used to determine whether a sample mean is significantly different from a hypothesized population mean when the population standard deviation is **unknown**.
Instead of using the population standard deviation σ, we use the sample standard deviation s.

The one-sample t-test formula is:

$$\boxed{t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}}$$

When to Use a T-Test?

A t-test is commonly used when:

- Population standard deviation is unknown.
- Sample standard deviation is used instead.
- Data is approximately normally distributed, especially for small samples.

The t-distribution is particularly useful for small samples because it accounts for the extra uncertainty introduced when estimating the population standard deviation from the sample.

Degrees of Freedom

For a one-sample t-test:

$$\boxed{df = n - 1}$$

Where:

- df = Degrees of freedom
- n = Sample size

Example

If:

$$n = 10$$

Then:

$$df = 10 - 1 = 9$$

The degrees of freedom are used to determine the appropriate t-distribution and critical value.

Example

Problem

A company claims that the average battery life of its product is **10 hours**.

A sample of 10 batteries gives:

$$\bar{x} = 9.5$$

Sample standard deviation:

$$s = 1$$

Test whether the average battery life is different from 10 hours at a 5% significance level.

Step 1: State Hypotheses

Because we want to check whether the mean is different:

$$H_0 : \mu = 10$$

$$H_a : \mu \neq 10$$

This is a two-tailed test.

Step 2: Significance Level

$$\alpha = 0.05$$

Step 3: Calculate Degrees of Freedom

$$df = n - 1$$

$$df = 10 - 1 = 9$$

Step 4: Calculate t-statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Substitute:

$$t = \frac{9.5 - 10}{1/\sqrt{10}}$$

$$t \approx -1.58$$

Step 5: Critical Value

For:

$$\alpha = 0.05$$

and:

$$df = 9$$

For a two-tailed test:

$$t_{critical} \approx \pm 2.262$$

Step 6: Compare

Our calculated value:

$$t = -1.58$$

Critical values:

$$-2.262, \quad +2.262$$

Since:

$$-2.262 < -1.58 < 2.262$$

the test statistic does not fall in the rejection region.

Therefore:

$$\boxed{\text{Fail to reject } H_0}$$

Step 7: Conclusion

> At the 5% significance level, there is not enough statistical evidence to conclude that the average battery life is different from 10 hours.